MIL-T-46586C (AR)
13 JUNE 1978
SUPERSEDING
MIL-T-46586B (MU)
29 February 1968

MILITARY SPECIFICATION

TUBE, IGNITER FOR CHARGE, PROPELLING, 175MM, M86A2
(CELLULAR POLYURETHANE)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication for one type of igniter tube designated as the Bell Igniter Tube for the 175MM, M86A2.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents.-The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-I-45607 - Inspection Equipment, Supply and Maintenance Of

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1376

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DRAWINGS (see 6.5)

ARRADCOM

8864105 - Bell Igniter Tube

(Copies of specification, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Igniter tubes.-The igniter tubes shall comply with all requirements specified on Drawing (Dwg.) 8864105 and with all requirements specified in applicable specifications and standards.

3.3 Density.-The density shall be 17.5 plus or minus 1.5 pounds per cubic foot throughout the length of the tube when tested as specified in 4.5.1.

3.4 Burning characteristics.-The polyurethane material shall start to disintegrate in 8 minutes and shall be completely consumed in 15 minutes when tested as specified in 4.5.2.

3.5 Tube condition.-The tube shall not have splits greater than 0.25 inch long in bell area extending completely thru the tube wall and terminating in the open end of the bell. There shall not be cavitations or surface defects greater than 0.25 square inch in bell area. All straight ends shall not have surface defects greater than 0.50 square inch.

3.6 First Article Inspection.-This specification contains technical provisions for the first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship.-All parts shall be free of dirt, grease, oil and other foreign material except that traces of mold release agent will be permitted. Tubes shall not have cracks of any size extending completely thru tube walls, and be free of holes (other than drawing requirements) extending thru the wall of any size caused by damage or molding flaws. Slick portions (skinning) of surface area may be missing as long as

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wall thickness requirements are met. Pits less than 0.25 inch in diameter and less than 1/16 inch in depth will be allowed. Unwetted areas of scrim shall not exceed one square inch. Scrim pattern must be visible over at least 50% of outside surface, and scrim frayed ends shall not exceed 1/16 inch except in holes. Surface porosity is acceptable if sections meet density testing requirements, see 4.4.3.1. All work performed shall be performed in a thorough workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48076 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Bell Igniter Tube	9864103	25

4.3.2 Inspections to be performed.-See MIL-A-48076 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48076.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Bell Igniter Tube				8864105
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Bell Igniter Tube Examination for Defects	25		3.2	4.4.2.1

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Bell Igniter Tube				8864105
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter of tube, min.		0.40%	3.2	Gage
102	Inside diameter of bell, max.		0.40%	3.2	Gage
103	Location of holes		0.40%	3.2	Gage
104	Hole missing		0.40%	3.2	Visual
105	Damage to the extent that part is rendered ineffective for subsequent use		0.40%	3.5	Visual
106	Density		Note a	3.3	4.5.1
107	Burning characteristic		Note b	3.4	4.5.2
108	Scrim missing in bell area		0.40%	3.2	Visual
Minor					
201	Outside diameter of tube, max.		0.65%	3.2	Gage
202	Outside diameter of bell		0.65%	3.2	Gage
203	Wall thickness		0.65%	3.2	Gage
204	Total length		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: a. See 4.4.3.1
b. See 4.4.3.2

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Density.-Five tubes from each lot shall be selected for this test. A specimen, between two and three inches in length, shall be cut from each end and one from the center (three specimens from each tube). The center, of the specimen cut from the center of the tube, must be within 1/2 inch maximum from the center of the tube from which it was cut. If any specimen fails to comply with the requirement of 3.3 the lot shall be rejected.

4.4.3.2 Burning characteristics.-Ten specimens, two from each of the tubes used in the density shall be used for this test. Each specimen shall weigh 5 plus or minus 1 gram. If any specimen fails to comply with the requirements of 3.4 the lot shall be rejected.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Test Methods and Procedures

4.5.1 Density.-A cylindrical section of tube 2 1/2 plus or minus 1/2 inch in length (three specimens from each sample tube) shall be weighed in air and then weighed in distilled water with a weight attached. The weight used shall also be weighed separately in distilled water. The temperature of the water shall be 23 plus or minus 1 degree Celsius and a small amount of wetting agent may be added. The specific gravity shall be calculated as follows:

$$\text{Specific gravity} = \frac{A}{A - (B-C)}$$

where:

A = weight of specimen in air
B = weight of specimen plus weight in water
C = weight of weight in water

The density shall be calculated as follows:

$$\text{Density } (gm/cm^3) = \text{specific gravity} \times 62.43$$

4.5.2 Burning characteristics.-Use two specimens from each of the sample tubes. Each specimen shall weight 5 plus or minus 1 gram. The specimen shall be placed in a muffle furnace maintained at a temperature of 1000 degrees plus or minus 50 degrees Celsius. Observations shall be made at one

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minute intervals until the specimen starts to disintegrate and every two minutes thereafter until the sample is completely consumed.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking

5.1.1 Level C.-Packing and marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Intended use.-The components covered by this specification are intended for use on the 175MM, M86A2.

6.2 Ordering Data.-See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals - See MIL-A-48078.-Submit equipment designs, as required, to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.5 Drawings.-Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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